



## A comparison of the immunostimulatory effects of the medicinal herbs *Echinacea*, Ashwagandha and Brahmi

Koji Yamada<sup>a</sup>, Pham Hung<sup>a</sup>, Tae Kyu Park<sup>c</sup>, Pyo Jam Park<sup>c</sup>, Beong Ou Lim<sup>b,\*</sup>

<sup>a</sup> Bioscience and Biotechnology, Faculty of Agriculture, Kyushu University, 6-10-1 Hakozaki, Fukuoka 812-8581, Japan

<sup>b</sup> College of Biomedical & Health Science, Department of Applied Biochemistry, Konkuk University, 322 Danwol-dong, Chungju, Chungbuk 380-701, Republic of Korea

<sup>c</sup> College of Biomedical & Health Science, Department of Life Science, Konkuk University, 322 Danwol-dong, Chungju, Chungbuk 380-701, Republic of Korea

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### ABSTRACT

**Aim of the study:** Herbs, as food or medicine, can strengthen the body and increase its resistance to illnesses by acting on various components of the immune system. For example, *Echinacea* is noted for its ability to enhance immune function, primarily through activation of the innate immune responses. Here, we investigated the potential for two herbs commonly found in India, Ashwagandha (*Withania somnifera*) and Brahmi (*Bacopa monnieri*), to enhance immune function and compared their effects to that of *Echinacea*.

**Materials and methods:** Sprague Dawley rats were fed a diet supplemented with 1% (w/w) *Echinacea*, Ashwagandha, or Brahmi for 4 weeks to examine their effects on immune function.

**Results:** The Brahmi diet stimulated more secretion of IgA and IgG in the serum compared to *Echinacea* or Ashwagandha. Whether or not lectin was present in the diet, the production of IgA, IgG and IgM in spleen lymphocytes increased with herbal supplements. The concentrations of IFN- $\gamma$  and IL-2 treated with LPS and ConA were significantly higher in the dietary herb than in the control. On the contrary, TNF- $\alpha$  production in rats receiving dietary herbal supplements was significantly lower compared to the control animals.

**Conclusion:** Herbal remedies based on *Echinacea*, Brahmi, or Ashwagandha can enhance immune function by increasing immunoglobulin production. Furthermore, these herbal medicines might regulate antibody production by augmenting both Th1 and Th2 cytokine production.

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### 1. Introduction

For many centuries, medicinal herbs have been used to treat diseases ranging from autoimmune diseases to cancer (Cundell et al., 2003; Kusunoki et al., 2004; Cassileth et al., 2008; Razaki et al., 2009). Animal experiments and modern clinical trials have shown that many of these herbs have immunomodulatory properties (Yale and Liu, 2004; Wang et al., 2008). Among the herbs, *Echinacea*, a native of North America, is widely used to prevent or to provide early treatment for the common cold (Blumenthal, 2004). Studies with *Echinacea* lend plausibility to the idea that herbs somehow stimulate the immune system and numerous clinical trials have documented the beneficial effects of *Echinacea* preparations.

The ancient Ayurvedic medicinal system of India is based on herbal remedies used to treat various symptoms and conditions. *Withania somnifera*, commonly known as Ashwagandha, has been

studied to some extent as an antioxidant (Rajasankar et al., 2009), but its effects on the immune system are poor (Malik et al., 2007; Khan et al., 2009). *Bacopa monnieri*, also known as Brahmi, is a traditional treatment for epilepsy and asthma and has antioxidant properties as it impairs the oxidation of fats in the bloodstream (Rajani et al., 2004). The application of biologically active new compounds derived from Ashwagandha and Brahmi extracts might make it possible to prevent or treat various immunological diseases.

Among diseases related to immune function, the prevention of infectious diseases and allergies (Mishra et al., 2000; Hofmann and Abraham, 2009), either by stimulation of defensive immunity or suppression of hypersensitivity, is essential. For many persons, the regression of defensive immunity is a major problem. Thus, the stimulation of immunological activity by herbal remedies such as *Echinacea*, Ashwagandha and Brahmi should be considered as new forms of treatment. However, information on the immune function and responsiveness to herbal components is limited. Previously, we reported that functional components such as the flavonoids, herb and dietary fibers regulate serum immunoglobulin (Ig) levels and Ig productivity of spleen and mesenteric lymph node lymphocytes in rats (Lee et al., 2007; Lim et al., 2007; Choi et al., 2010). In the

\* Corresponding author. Tel.: +82 43 840 3570; fax: +82 43 851 4169.

E-mail address: [beongou@kku.ac.kr](mailto:beongou@kku.ac.kr) (B.O. Lim).

**Table 1**  
Compositions of experimental diets.

| Ingredients            | Diet composition (%) |                  |             |        |
|------------------------|----------------------|------------------|-------------|--------|
|                        | Control              | <i>Echinacea</i> | Ashwagandha | Brahmi |
| Herbal extract         | 0                    | 1                | 1           | 1      |
| Cornstarch             | 37                   | 36               | 36          | 36     |
| Casein                 | 20                   | 20               | 20          | 20     |
| Dex-cornstarch         | 13                   | 13               | 13          | 13     |
| Sucrose                | 10                   | 10               | 10          | 10     |
| Fiber (cellulose)      | 5                    | 5                | 5           | 5      |
| Soybean oil            | 10                   | 10               | 10          | 10     |
| Mineral mix            | 3.5                  | 3.5              | 3.5         | 3.5    |
| Vitamin mix            | 1                    | 1                | 1           | 1      |
| L-Cystine              | 0.3                  | 0.3              | 0.3         | 0.3    |
| Choline bitartrate     | 0.25                 | 0.25             | 0.25        | 0.25   |
| Tert-butylhydroquinone | 0.0014               | 0.0014           | 0.0014      | 0.0014 |

present study, we examined the effect of dietary herb on Ig and cytokine production in the rat spleen lymphocytes.

## 2. Materials and methods

### 2.1. Materials

Concanavalin A and lipopolysaccharide were purchased from List Biological Labs (Campbell, CA, USA) and dissolved in PBS (pH 7.4) for cell culture experiments. ELISA assays to detect IL-2, IFN- $\gamma$ , TNF- $\alpha$ , IL-4 and IL-6 were from Endogen (Woburn, MA, USA) and ELISA assays to detect IgA, IgG and IgM were from Bethyl Laboratories (Montgomery, TX, USA).

### 2.2. Animal and diets

All animal care techniques were performed within the guidelines approved by the Institutional Animal Care and Use Committee. Four-week-old male Sprague Dawley rats were housed individually in a windowless room with controlled temperature and light (20–23 °C and light cycle of 08:00–20:00 h). After acclimation for one week, the rats were divided into four groups of ten rats and were given free access to experimental diet. The experimental diets were manufactured according to the AIN-93G standard (Reeves et al., 1993) and contained 1% (w/w) dietary herb (Table 1) as follows: 10 g of *Echinacea* per kg of rat feed, 10 g of Ashwagandha per kg of rat feed, and 10 g of Brahmi per kg of rat feed. Herbal extract of *Echinacea purpurea* was from Api Ltd. (Gifu, Japan), extract prepared from *Withania somnifera* (Ashwagandha) containing 1.1% alkaloids and 3.6% withanoids was from Gwalior Ltd. (Gwalior, India), and extract prepared from *Bacopa monniera* Linn. (Brahmi) containing 12.8% saponin was from Gwalior Ltd. (Gwalior, India).

Body weight and food intake were recorded every other day. After four weeks of feeding, rats were sacrificed by withdrawing

blood from the abdominal aorta under light diethyl ether anesthesia. Serum Ig concentration was measured by ELISA (R&D Systems, Inc., Minneapolis, MN, USA).

### 2.3. Preparation of spleen lymphocytes

The spleen lymphocytes isolated from each dietary group were cultured in 100 ml FBS/RPMI1640 medium. The spleen lymphocytes were adjusted to  $2 \times 10^6$  cells/ml in 24 well microtiter plates in a final volume of 1 ml. Cells were incubated at 37 °C for 48 h in the absence or presence of 10  $\mu$ g/ml concanavalin A (ConA) and lipopolysaccharides (LPS). The concentration of IgA, IgG and IgM was measured by ELISA (R&D Systems, Inc., Minneapolis, MN, USA), as specified by the manufacturer. The concentration of IL-2, IFN- $\gamma$ , TNF- $\alpha$ , IL-4 and IL-6 was also measured by ELISA (R&D Systems, Inc., Minneapolis, MN, USA) as specified by the manufacturer.

### 2.4. Statistical analysis

Data were analyzed using SAS software (SAS Institute, Cary, NC, USA) by Duncan's new multiple range test to evaluate the exact nature of the relationships among the groups (Duncan, 1955).

## 3. Results

### 3.1. Growth variables and serum lipid level

The effect of the dietary herbs on food intake and weight gain in the Sprague Dawley rats was measured (Table 2). In the Brahmi diet, a tendency to ingest less food was observed and there was a slight decrease in body weight compared to the control rats and *Echinacea*-fed rats. In the Ashwagandha diet, weight gain was slight higher than in the control or *Echinacea* groups, although the difference was not significant. In the case of tissue weight, significant

**Table 2**  
Effect of dietary herb on the body weight, tissue weight and serum lipid levels of Sprague–Dawley rats.

|                               | Control                       | <i>Echinacea</i>              | Ashwagandha                   | Brahmi                        |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| Initial body weight (g)       | 121.0 $\pm$ 4.1               | 122.1 $\pm$ 1.8               | 120.5 $\pm$ 3.2               | 121.0 $\pm$ 3.5               |
| Weight gain (g)               | 189.9 $\pm$ 9.7               | 189.5 $\pm$ 4.7               | 204.1 $\pm$ 3.5               | 182.7 $\pm$ 8.6               |
| Food intake (g/day)           | 17.9 $\pm$ 0.86 <sup>ab</sup> | 18.2 $\pm$ 0.42 <sup>ab</sup> | 18.7 $\pm$ 0.37 <sup>a</sup>  | 16.5 $\pm$ 0.68 <sup>b</sup>  |
| Food efficiency (diet/weight) | 2.6 $\pm$ 9.05                | 2.7 $\pm$ 0.04                | 2.6 $\pm$ 0.04                | 2.5 $\pm$ 0.06                |
| Tissue weight                 |                               |                               |                               |                               |
| Spleen                        | 0.58 $\pm$ 0.03               | 0.56 $\pm$ 0.03               | 0.58 $\pm$ 0.04               | 0.61 $\pm$ 0.04               |
| Liver                         | 8.72 $\pm$ 0.41 <sup>a</sup>  | 9.21 $\pm$ 0.14 <sup>b</sup>  | 9.87 $\pm$ 0.47 <sup>b</sup>  | 9.41 $\pm$ 0.20 <sup>b</sup>  |
| Lung                          | 1.41 $\pm$ 0.05               | 1.51 $\pm$ 0.09               | 1.46 $\pm$ 0.07               | 1.38 $\pm$ 0.07               |
| Kidney                        | 1.83 $\pm$ 0.08 <sup>a</sup>  | 1.98 $\pm$ 0.06 <sup>ab</sup> | 2.08 $\pm$ 0.06 <sup>b</sup>  | 1.92 $\pm$ 0.06 <sup>ab</sup> |
| Heart                         | 1.10 $\pm$ 0.05               | 1.12 $\pm$ 0.03               | 1.12 $\pm$ 0.05               | 1.04 $\pm$ 0.04               |
| Adipose                       | 10.07 $\pm$ 1.39 <sup>a</sup> | 10.76 $\pm$ 0.75 <sup>a</sup> | 11.89 $\pm$ 0.56 <sup>a</sup> | 7.08 $\pm$ 0.94 <sup>b</sup>  |

Each data value is the mean  $\pm$  SE ( $n = 10$ ), and values not sharing a common superscript letter are significantly different at  $p < 0.05$ .

**Table 3**

Effect of dietary herb on immunoglobulin in serum of Sprague–Dawley rats.

| Serum       | Control                  | <i>Echinacea</i>         | Ashwagandha              | Brahmi                   |
|-------------|--------------------------|--------------------------|--------------------------|--------------------------|
| IgA (μg/ml) | 26.7 ± 0.69 <sup>a</sup> | 26.8 ± 1.05 <sup>a</sup> | 24.1 ± 2.14 <sup>a</sup> | 35.5 ± 3.88 <sup>b</sup> |
| IgG (mg/ml) | 0.75 ± 0.13 <sup>a</sup> | 0.77 ± 0.15 <sup>a</sup> | 0.81 ± 0.17 <sup>a</sup> | 1.56 ± 0.15 <sup>b</sup> |
| IgM (μg/ml) | 343.1 ± 17.11            | 335.1 ± 20.61            | 389.2 ± 74.86            | 398.5 ± 34.33            |
| IgE (ng/ml) | 12.5 ± 0.93              | 12.5 ± 1.46              | 11.1 ± 1.03              | 13.4 ± 0.97              |

Data are means ± SE (n = 5). (a–b) Values not sharing a common superscript letter are significantly different at  $p < 0.05$ .

effects were observed in the kidney and liver adipose tissue due to the herbal diets. Kidney weight was significantly higher in the Ashwagandha diet than in the control, and in the liver, there was a progressive increase in weight gain in the rats fed *Echinacea*, Ashwagandha, and Brahmi, respectively. Therefore, the effect of the dietary herbs on weight gain in the kidney and liver adipose tissue are different.

### 3.2. Serum and spleen lymphocyte immunoglobulin production

We determined production of the immunoglobulins in the serum and in the splenic lymphocytes. The IgA and IgG concentration in the serum of rats on the Brahmi diet was significantly higher compared to the *Echinacea* and Ashwagandha diets, although the IgE and IgM concentrations did not differ significantly among the groups (Table 3). Next, we looked at the effect of the dietary herbs on immunoglobulin production by splenic lymphocytes to evaluate the response of the systemic immune system to the dietary herbs. We found that IgA, IgG, and IgM production in rats fed a diet supplemented with the herbs was significantly higher than in the control animals (Table 4). Likewise, all three dietary herbs were found to stimulate IgA, IgG, and IgM production in the presence of LPS or ConA (Table 4).

### 3.3. Cytokine production by spleen lymphocytes

In the absence of lectins, the dietary herbs significantly increased IL-2 and IFN-γ secretion (Fig. 1A). In the presence of LPS, both *Echinacea* and Brahmi stimulated significantly higher IL-2 secretion and IFN-γ compared to the control and Ashwagandha diets. In contrast, the Ashwagandha diet did not significantly influence the production of IFN-γ and IL-2. ConA-induced IL-2 and IFN-γ production in rats fed dietary herbs was significantly higher compared to the control diet (Fig. 1A). In the absence of ConA, the TNF-α concentration in the spleen lymphocytes was lower in *Echinacea*-fed rats compared to the other diets (Fig. 1A). When the spleen cells were cultured with LPS and ConA, a similar response pat-

tern was observed. Feeding of dietary herb was accompanied by a disturbance in the IL-4 and IL-6 levels (Fig. 1B). With or without LPS treatment, IL-4 production did not differ among the groups (Fig. 1B). However, LPS and ConA-induced IL-6 production in the Brahmi diet was significantly higher compared to the other diets (Fig. 1B).

## 4. Discussion

The immune system is an important defensive system to protect the human body from invasion by various pathogens such as allergens, viruses and bacteria (Chaplin, 2010). In particular, IgA plays a major role in mediating allergy while IgG and IgM are thought to regulate allergy by preventing the binding of allergen. We observed that IgA, IgG, and IgM production was significantly higher in the splenic lymphocytes of rats fed *Echinacea*, Ashwagandha and Brahmi than in the rats fed a control diet (Table 4). When cells were cultured with ConA and LPS, a similarly response pattern observed (Table 4). Our results suggest that the dietary herbs stimulate IgA, IgG and IgM production by rat splenic lymphocytes in the absence or presence of lectin. Thus, dietary herbs might alleviate inflammatory reactions in the systemic immune system of humans.

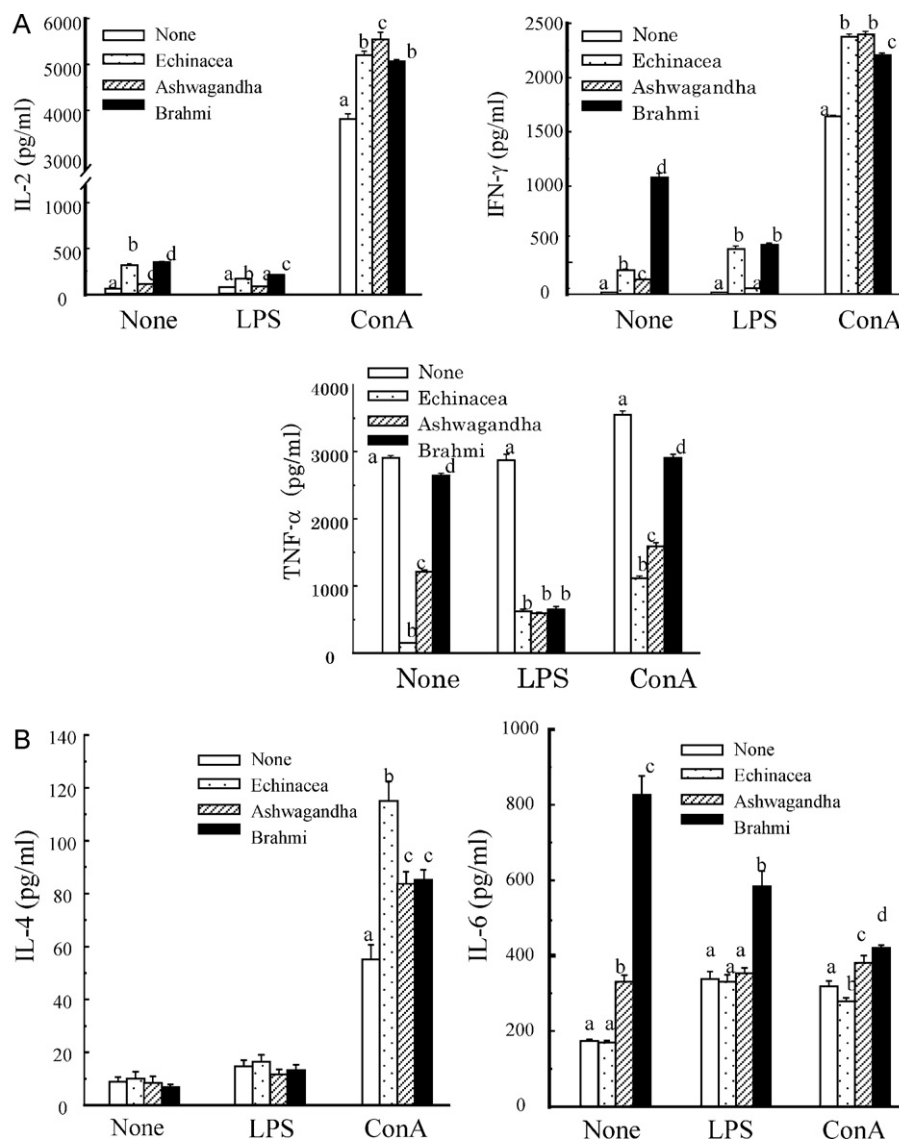
Immunoglobulin production is class-specifically regulated by various types of lymphokines (Iemoli et al., 2010). In addition to ConA and LPS, several T and B cell mitogens have been used in immunologic studies. Because some of these mitogens may stimulate selected lymphocytes populations, we used ConA and LPS, both which stimulates Ig production by splenic lymphocytes in culture (Lim et al., 2003a, 2003b) and are inducers of the production of these cytokines (Lim et al., 2004). The cytokine network plays an important role in the inflammatory and immune responses (Pandey et al., 2009; Bak and Mikkelsen, 2010). We demonstrated that administration of dietary herbs might modulate cytokine production in splenic lymphocytes. Of the dietary herbs examined, *Echinacea* and Brahmi stimulated IFN-γ production to the greatest extent. In the presence of ConA, the spleen cells from rats fed dietary herbs produced fairly high levels of IFN-γ. Similarly, a significant increase in

**Table 4**

Effect of dietary herbs on immunoglobulin production by spleen lymphocytes.

| Diet groups      | Mitogens (10 μg/ml) |      | Ig concentrations (ng/ml) |                           |                             |
|------------------|---------------------|------|---------------------------|---------------------------|-----------------------------|
|                  | LPS                 | ConA | IgA                       | IgG                       | IgM                         |
| Control          | –                   | –    | 20.9 ± 0.41 <sup>a</sup>  | 9.6 ± 0.85 <sup>a</sup>   | 160.7 ± 2.49 <sup>a</sup>   |
| <i>Echinacea</i> | –                   | –    | 106.6 ± 1.34 <sup>b</sup> | 20.5 ± 0.88 <sup>b</sup>  | 658.6 ± 9.60 <sup>b</sup>   |
| Ashwagandha      | –                   | –    | 108.1 ± 2.49 <sup>b</sup> | 44.6 ± 0.98 <sup>c</sup>  | 786.5 ± 22.01 <sup>c</sup>  |
| Brahmi           | –                   | –    | 101.4 ± 1.68 <sup>c</sup> | 54.3 ± 1.71 <sup>d</sup>  | 601.9 ± 13.92 <sup>bc</sup> |
| Control          | +                   | –    | 18.0 ± 1.16 <sup>a</sup>  | 9.2 ± 0.50 <sup>a</sup>   | 171.1 ± 8.24 <sup>a</sup>   |
| <i>Echinacea</i> | +                   | –    | 191.9 ± 1.46 <sup>b</sup> | 18.5 ± 0.76 <sup>b</sup>  | 1745.5 ± 14.11 <sup>b</sup> |
| Ashwagandha      | +                   | –    | 204.2 ± 3.94 <sup>c</sup> | 41.9 ± 1.06 <sup>c</sup>  | 2118.7 ± 32.64 <sup>b</sup> |
| Brahmi           | +                   | –    | 81.4 ± 1.70 <sup>d</sup>  | 35.9 ± 0.85 <sup>cd</sup> | 663.4 ± 6.59 <sup>c</sup>   |
| Control          | –                   | +    | 9.9 ± 0.76 <sup>a</sup>   | 7.2 ± 0.45 <sup>a</sup>   | 79.3 ± 2.89 <sup>a</sup>    |
| <i>Echinacea</i> | –                   | +    | 78.4 ± 0.92 <sup>b</sup>  | 5.1 ± 0.22 <sup>b</sup>   | 642.9 ± 7.51 <sup>b</sup>   |
| Ashwagandha      | –                   | +    | 80.2 ± 1.51 <sup>b</sup>  | 10.4 ± 0.41 <sup>c</sup>  | 525.4 ± 11.86 <sup>c</sup>  |
| Brahmi           | –                   | +    | 101.5 ± 2.62 <sup>c</sup> | 18.7 ± 0.75 <sup>d</sup>  | 806.1 ± 20.10 <sup>d</sup>  |

Lymphocytes were cultured with or without ConA and LPS for 48 h. Culture supernatants were then collected to measure Ig contents. Data are means ± SE (n = 5). (a–d) Values not sharing a common superscript letter are significantly different at  $p < 0.05$ .



**Fig. 1.** (A and B) Activation-induced lymphocytes cytokine secretion in spleen lymphocytes of SD rats fed dietary herb. Lymphocytes were cultured with or without LPS and ConA for 48 h. Culture supernatants were then collected to measure cytokines. Data are means  $\pm$  SE ( $n=5$ ). Values not sharing a common superscript letter are significantly different at  $p < 0.05$ .

IL-2 productivity was observed in spleen lymphocytes stimulated by LPS from those fed with dietary herb.

We reported previously that the effect of dietary fiber was exerted through the Th1 cell activity (Yamada et al., 2003). Here, we report that the effect of dietary herbs on IFN- $\gamma$  and IL-2 was also strong. We saw increased cytokine production from Th1 cells by treating the cells with dietary herbs, even after persistent inflammation. These data suggest that treating ConA- and LPS-induced lymphocytes with dietary herb could strengthen the immune system by regulating cytokines such as IL-2 and IFN- $\gamma$ . Regulation of the production of the Th2-associated cytokines, IL-4 and IL-6, by administration of dietary herb suggests that these herbs might be inducing a balanced differentiation of the spleen lymphocytes toward the production of both Th1 and Th2 cytokines. The IL-6 concentration was higher in the Brahmi-treated lymphocytes stimulated with ConA and LPS than in the *Echinacea* and *Ashwagandha*-treated groups. IL-6 is a pleiotropic cytokine influencing the antigen-specific immune response (Artym et al., 2004) and inflammatory reaction (Neveu et al., 2010). Because IL-6 is one of the major factors for the terminal differentiation of activated B cells into Ig-producing cells (Hori et al., 1988), there is a possibility that

IL-6 might be involved in enhancing Ig production by B cells upon Brahmi treatment.

In our study, the administration of *Echinacea* and Brahmi induced significant increases in the serum IgG level and all Ig production in the spleen. Furthermore our study suggests the possibility of clinical applications of dietary herb containing Brahmi based on its immunoregulatory effects. Although the mechanism by which the herbal components modify immune indices is not yet known, our results point to a new possibility for the pharmacological role of dietary herbs.

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